Guided Flow Matching for Extreme Weather Simulation

Master's Thesis

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Abstract

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Chapter 1

Introduction

Machine learning models have reached state-of-the-art skill in weather forecasting, where they now rival traditional physics-based solvers at a fraction of their computational cost. One class among them are generative probabilistic models. First developed in the image domain, they have recently been carried over to weather. There, they now define the state of the art in probabilistic forecasting.

In the image domain, the applications of interest almost all *condition* a pretrained model rather than sample from it unconditionally: text-to-image generation, editing, and inpainting each steer the generative process towards a desired target through *guidance*. In the weather domain, by contrast, generative models are mostly used in their unguided form, as residual predictors that both overcome the oversmoothing of deterministic models and enable ensemble forecasts out of the box. The use of guidance in the weather domain remains largely unexplored. The few exceptions steer forecasts towards specific extreme events: [cite] generate windstorms, [cite] minimise precipitation, and [cite] produce heatwaves.

Extreme events cause a disproportionate share of weather-related damage, yet their rarity makes them precisely the cases in which forecast ensembles are thinnest and observational data for study scarcest. This motivates our central question: can we deliberately generate realistic extreme weather trajectories with a generative forecasting model, rather than wait for an unguided ensemble to sample them? Building on the recent works above, we tackle the question by splitting it into three subproblems—specifying the event, guiding the model, and evaluating the result—which together form a general framework to *specify*, *generate*, and *evaluate* extreme events with generative weather models.

We build on ArchesWeather, a residual flow-matching model, and focus throughout on surface-temperature heatwaves over a chosen region of interest; the methodology itself is model and target agnostic. The three subproblems are also our three contributions:

1. **Specify.** A heuristic that uses historical states consistent with a given start state to define a plausible target trajectory for the region and variable of interest.
2. **Guide.** A gradient-based guidance method, adapted from classifier guidance, that steers the forecast onto the specified target by driving the gap to zero rather than by tuning guidance strength for a visually plausible result, thus requiring minimal tuning and minimising the intervention.
3. **Evaluate.** A set of flow and weather plausibility tests. In particular, for output plausibility we propose a benchmark, built from the same historical data, that tests generated trajectories for spatial, pressure-level, and cross-variable coherence, using the historical states as anchors and the unguided forecast as baseline.

The remainder of the thesis is organised as follows. Chapter 2 introduces the technical background on flow matching, guidance, generative weather models, and seminal applications of guidance in the weather domain. Chapters 3 to 5 develop the three phases of the framework in turn: specifying the target event, designing the guidance, and defining the evaluation. Chapter 6 presents the experiments and Chapter 7 discusses them. Chapter 8 concludes with a summary, directions for future work, and final remarks.

Chapter 2

Background

2.1 Flow matching

Flow matching [2] models a data distribution p_1 as the endpoint of a continuous transport of a tractable prior $p_0 = \mathcal{N}(0, I)$. Let $(p_t)_{t\in[0,1]}$ be a *probability path* with these endpoints, generated by a time-dependent *velocity field* v_t through the flow

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\phi_t(x) = v_t\big(\phi_t(x)\big), \qquad \phi_0 = \mathrm{id}, \tag{2.1}$$

whose push-forward carries p_0 to p_t; equivalently, v_t and p_t obey the continuity equation $\partial_t p_t + \nabla\cdot(p_t\, v_t) = 0$. Sampling then reduces to drawing $x_0 \sim p_0$ and integrating the flow to $t = 1$.

Training regresses a network v_θ onto the field that generates p_t. This marginal field is intractable, but it decomposes as an expectation of *conditional* fields, $v_t(x) = \mathbb{E}_{x_1\sim p_{1|t}(\cdot\,|x)}\big[\, v_t(x \mid x_1)\,\big]$, each $v_t(\cdot \mid x_1)$ generating a simple path from p_0 to a fixed datum x_1. Flow matching exploits the fact that the tractable *conditional* objective

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\, x_1\sim p_1,\, x_t\sim p_t(\cdot\,|x_1)} \Big\| v_\theta(x_t, t) - v_t(x_t \mid x_1) \Big\|^2 \tag{2.2}$$

shares its gradient with the intractable marginal one [2]. Under the linear (optimal-transport) path $x_t = (1-t)\,x_0 + t\,x_1$ with $x_0 \sim p_0$, the conditional field is the constant displacement $v_t(x_t \mid x_1) = x_1 - x_0$, so training amounts to least-squares regression onto $x_1 - x_0$.

At inference we draw $x_0 \sim \mathcal{N}(0, I)$ and integrate $\dot{x}_t = v_\theta(x_t, t)$ with a few Euler steps; independent draws yield independent samples. Since this map from x_0 to x_1 is a single deterministic and differentiable trajectory, its output can be steered by differentiating a target loss through the flow—the mechanism on which our guidance builds (Chapter 4).

2.2 Guidance methods

Guidance methods were originally develped in the context of generative models based on diffusion. However, through the translation formulas from diffusion to flow matching we can easily identify their flow matching counterparts.

Classifier guidance

2.2.1 Backpropagating through the ODE

FlowGrad

D-Flow

Control paper

FlowGrad [3] backpropagates such a gradient through the sampling trajectory; D-Flow [1] instead optimises the source x_0, which for Gaussian paths projects the update onto the data manifold; and Wang et al. [4] frame the steering as optimal control. Our method adopts this gradient-based view but, rather than tuning a guidance strength, drives an explicit target gap to zero (Chapter 4).

2.2.2 Classifier guidance

Classifier free guidance ...

We can translate the derived formula to flow land using following relation: ...

The general algorithm of classifier guidance works as follows.

...

A key finding is that in order to make guidance actually work, we have to abandon probability land, by increasing the strength of the guidance. In fact, when we start to play with *lambda*, the guided output is not a sample of the conditional probability distribution anymore.

2.2.3 Guidance schedules

The quest for the "best schedule" is still something that hasn't been solved in the guidance literature, although many heuristics have been proposed.

2.3 Generative weather models

Since the start of the machine learning weather literature a few models based on generative architectures have been proposed. Among the most prominent ones we can find GenCast.

GenCast ... operational ...

To prototype our method however the computational and memory requirements for GenCast are quite elevate. For this reason we pick as model of choice ArchesWeather.

2.4 ArchesWeather

We now spend some time presenting ArchesWeather in detail. This will enable us to introduce the relevant pieces we will be working on.

In our setting the model is a *residual* predictor: the generated x_1 is an increment added to a base state (a previous state or a deterministic forecast), so each ensemble member is that base plus one integrated trajectory (Chapter 4).

2.4.1 Dataset

2.4.2 Architecture

2.4.3 Training

2.4.4 Evaluation

2.5 Guidance in weather modeling

The last piece of the puzzle before presenting our approach is to expose what people have alredy tried out, enabling us to expose the relevant gaps that we want to fill.

Digging into paper archives, I was able to find only three notable papers that were approaching the same general task as us.

The first one is ...

The most similar in nature, and the one from which we will blatently borrow some aspects of the evaluation, is also the one that approaches guidance and its components the most thorrouwly.

We have thus presented all relevant background pieces for the exposition of our contributions. We will spend the next three chapters introducing each of these key aspects.

Ich habe leider vergessen, den Dauerauftrag für die Wohnungsmiete in meinem E-Banking rechtzeitig zu stoppen. Könnten Sie mir den bereits überwiesenen Mietbetrag für Juli bitte zurückerstatten? Die Zahlungsbestätigung habe ich Ihnen im Anhang beigefügt. Vielen Dank im Voraus. Freundliche Grüsse Alain

Chapter 3

Specifying events of interest

3.1 Focus events in this project

While we focus on surface temperature, the method we develop is general. However, in order to specify extreme events in other variable, we would require a higher expertise. So here we heavily focus on the methodology itself, leaving exploration of other variables to experts in the field.

It is really important to understand how to evaluate what we are doing here. Since we are aiming at a specific region of interest, and we are guiding only wrt the region of interest, we are loosing the global coherence of the weather states. We shall thus evaluate the ability of our method to generate locally coherent weather scenarios, over the region of interest.

3.2 Mask

The mask is a set of weigths (sum to 1) over a tensor having the same dimension as the weather states, non-zero at the variable and region of interest.

We experimented with a gaussian mask, but realized in our experiments that the best thing to do is equal weighting inside the region. The correct way to define the region of interest is to choose a greater mask size instead of using some weighting scheme fading out of an epicenter. The mask size should be chosen to cover the entire area of interest, ensuring that all relevant data is included in the analysis. A larger mask size can help to reduce noise and improve the accuracy of the results, while a smaller mask size may exclude important information. It is important to carefully consider the appropriate mask size for each specific application to achieve optimal results. Either way we (most probably) loose the global coherence for the guided weather states, so we might as well define directly the region of interest with a mask. Additionally, this enables us to actually make statements about "the region of interest", having equal weights everywhere. Introducing different weigths might bias the generation in a way that is not desired.

3.3 Target trajectory

In order to define plausible average trajectories of change, we define following procedure: 1. Select start state 2. Define mask 3. Retrieve historical states that are consistent with the start state and the mask (+- some tolerance) 4. Compute the distribution of average trajectories of change for the historical states 5. Filter a quantile of the distribution to define the target trajectory (e.g. 0.95 quantile for extreme events) 6. Define a gaussian distribution at each time step of the target trajectory, with mean equal to the target trajectory and variance equal to the variance of the historical states at that time step. 7. Sample enough trajectories from the gaussian distribution to define a set of target trajectories that will be used to guide the generative model. 8. Select the target trajectories of interest in terms of extremeness to build scenarios. 9. Save the trajectories and corresponding states used to define the sampling distribution. These will be used to evaluate the plausibility of the generated trajectories as means of comparison. 10. Run benchmark tests and evaluate.

This heuristic enables us to anchor the guidance target and define a benchmark at the same time. More concretly, we avoid the issue of having to tune the strength of the guidance to achieve a certain plausible result. We reverse the problem by defining the plausible result first and then designing an algorithm to achieve that result and checking that both the algorithm and its result are well behaved.

We explicitly define a heatwave in order to design guidance methods. This means that we decide to represent a specific subset of state trajectoriesin this work. However, our method is flexible to any kind of target: you can swap your definition of the event you want to replicate and use the guidance to generate that kind of event. This really only affects the target definition.

This choice enables us to side step the problem that guiding towards a classifier's gradients or minimizing / maximizing a certain variable and stopping when we achieved the wanted result. If we specify the carachteristic of the wanted result we side step this problem entirely. The problem then shifts to retrieve a representative set of the event we have in mind. This is however a requirement for the first methodology. So we could say that the problem shifts towards defining an appropriate algorithm that can achieve a representative state similar to the specified set. Making this step explicit directly enables us to then compare the results against this set, and characterize its flaws / benefits directly.

Chapter 4

Guidance design

4.1 General method

Building on the idea of classifier guidance we identify following

Assumption about using the gradient. We explicitly assume that in order to produce realistic result we have to tie the perturbations to the models' gradient. We use the none-gradient tied perturbations as baseline for all our experiments.

The hope is also that the gradient is informative in the sense that: if a method pushes a not secondary target variable to a certain distribution / average value, we would also expect that if we push this variable to the achieved average value, that the previously targeted variable would achieve the same result. This we explicitly test in test5.

The base guidance algorithm we use works as follows: ...

4.2 Guidance loss

The loss we use to guide the generative model is defined as follows: ...

One of the main contributions of this thesis is the realization that in the weather domain, we can define a guidance target, which can be achieved precisely. By precisely I mean that the loss can be driven to zero. While in the image domain, there is always a ground truth during training, at inference we play with different guidance strengths to identify a range where the generated images are good looking. In the weather domain however, we can define a target that can be closed by tuning the guidance strength.

This is great because instead of tuning the guidance strength to find a range of plausible results, we can define what plausible results look like a priori (guidance target sampling trajectory defined through historical data and plausibility tests), and optimize the guidance strength to achieve this target.

We essentially have a very similar setting to the guided diffusion for precipitation paper, but we make it more general in these ways:

- we define a guidance target explictly, in contrast to observe results across different lambdas
- we experiment with different a_t trajectories, in addition to guiding computing the gradient at the first step
- we test level of intensity by shifting the intenisty parameter from the flow (lambda) to the trajectory definition (phi_n)
- Our loss is driven to zero in terms of the difference from the target, instead of being minimized iteratively (a more explicit way to encode the target scenario)

4.3 Guidance algorithms

In this section we develop different variants of classifier free guidance, which we adapt to this setting. We explore the different variants, tune them, expose their flaws, assess their computational requirements, and compare their respective trade off. In the end we close in on the most promising method in terms of reliability and computational requirements, evaluated both in terms of theoretical cost, but also in terms of practical factors such as difficulty of tunining and potential need for multiple runs.

4.3.1 CLOSE-GAP

Greedy

Step is normalized to close the gap as prescribed by the guidance schedule.

4.3.2 OPTIMIZE-SCHEDULE

The step is allowed to take the gradient norm into consideration. The schedule is optimized to minimize the loss while also complying to a regularization term. Possible regularization terms include the total guidance. We can dream up arbitrary penalty terms such as angular defelction, ...

The problem of this optimization is that the gradient applyed for the guidance at each step changes with respect to the schedule. The optimization is therefore highly non-convex and very hard to solve. In fact, the optimizer will converge very fast to a local minimum, highly influenced by the initialization of the schedule.

LBFGS

4.3.3 OPTIMIZE-GAIN

Secant method

The step is allowed to take the gradient norm into consideration. Since the loss is almost linear with respect to the guidance strength, we can use a secant method to find the guidance strength that minimizes the loss.

In this case we define the secant method as follows:

4.3.4 Comparison

TABLE

Given the analyzed properties of the algorithms, in all our experiments we will make use of the latter presented method.

Chapter 5

Evaluation

To evaluate the performance of our guidance algorithm we need to test the two requirements we have set for it: extremeness and realism.

The first concern is whether the algorithm is able to achieve the target, and how well it does so. This entails testing the algorithm's ability to drive the gap to zero, and how well it does so across different settings. We will test the algorithm for convergence, stability, and trajectory deviation. These tests adress will help to unveil potential failure modes of the algorithm and showcase which algorithm design choices are more adequate for different settings.

The second concern is whether the generated states are realistic, in terms of temporal, spatial, vertical, and cross-variable coherence.

— redundant second sentence, to merge In our experiments we will test these properties analyzing the impact of the single parameters of our experimental setting. In particular we will test different modes or ranges for the single parameters and then assess these properties and report on them.

— ass covering part if I don't manage to perform the realism quantitative tests In this work we focus on designing a simple guidance method and empirically test its parameters and the properties of the results generated across different settings. We put less emphasis on the physical realism of the definition of the guidance target and the realism of the generated trajectories, and defer this to a separate line of work. Though, we have these two in mind when designing the guidance method throughout.

In addition to these tests, in the discussion we will also discuss different aspects of the results, such as the diversity of the generated trajectories, and the informativeness of the gradient, and other aspects that are not directly tested in the evaluation framework, but are of interest to the overall evaluation of the method and its potential applications.

5.1 Flow-time assessment

5.1.1 Parameters

In flow-time tests we assess:

- The algorithm variants
- Different guidance schedules

with the purpose of selecting the most promising method to perform the final experiments with.
The selection is based on the following criteria.

5.1.2 Tests

Before testiong whether our algorithm produces realistic results we have to check that it actually works. That is, it achieves the set target, it does so reliably across different settings, and it does so in a stable manner.
In particular we are interested in testing following:

- The algorithm converges reliably.
- The algorithm doesn't have an oscillatory behaviour in terms of the average trajectory, and in terms of the flow angles and magnitudes of the vector field. (the flow trajectory doesn't deviate too much from the unguided trajectory).

— merge the two paragraphs in a single enumeration, can then refer the test by numbers, give the test a clear identifier
— remove the across runs from the bullets We will tests these properties with following metrics across different runs:

- The gap to the target is driven to zero, and we report the average and std of the gap across different runs.
- The flow angles and magnitudes are reported across the flow steps, and we report the average and std of the angles and magnitudes across different runs.

- The flow trajectory is compared to the unguided trajectory in PCA latent space, and we report the average and std of the distance between the two trajectories across different runs.

Method selection criteria:

- Total amount of guidance applied across the flow steps
- The pushback from the unguided trajectory
- Computational and memory requirements
- The number of hyperparameters to tune, and the difficulty of tuning them
- The number of runs required to achieve the target across different settings
- The guidance method is robust to different settings, in the sense that it can achieve the target across different settings, and it does so in a stable manner.

It's here that we test the developed algorithms and their different variants across different parameter settings, and select the most promising one for the final experiments.

5.2 Weather time tests

5.2.1 Parameters

In an initial phase of the evaluation we want to test parameters in isolation, to both describe their properties, but also to select relevant settings for larger experiments. We can test these changes by combining them in bigger experiments, that enable us to use other parameters as control.

We start by testing different parameters of the region of interest such as mask centers and sizes across different start states. We then examine how reproducible are the generated target variable patterns by pushing its most correlated variable instead. We finally proceed to test intervention across different intensity profiles.

Throughout these initial experiments we will change start states while setting N=2 and M=3, such that we can control for effects produced by setting N=1, and effects related to the specific seed, while keeping computational and memory requirements lowest possible.

These experiments will help us understand the properties of the guidance algorithm we developed, and how its parameters influence the results. As for the guidance method, the described quantitative tests will help us in selecting parameters that produce the most desirable results under this lense.

Region of interest

- different sizes, same center
- different center, same size
- different regions across globe

Variable of interest

For reasons of simplicitly we explicitly only on heatwaves here. However, to assess the gradient informativeness we push also other variables in the attempt to reproduce the same pattern on the first experiment. This might also open up a way to produce more diverse results.

- Temperature first and then push corresponding top-k variables to previosly generated values.

Intervention profile

- Push to different intensities
- Assess what happens when you stop pushing (do states reconverge to unguided trajectory?)

5.2.2 Tests

Once we have selected the guidance method and its parameters, we want to assess the realism, in addition to other properties, of the generated trajectories. In particular we want to test the following properties.

Method tests

- The gradient is informative, in the sense that if we push a not secondary target variable to a certain distribution / average value, we would also expect that if we push this variable to the achieved average value, that the previously targeted variable would achieve the same result.

- Diversity test: the generated trajectories are diverse, in the sense that they are not all the same, and they cover a range of possible outcomes.

Realism tests

Realism tests are of high importance in this work, since we are explicitly guiding the generative model possibly outside of its training distribution. However, these tests are not particular to this work. In fact they concern the general problem of evaluating the coherence of data driven models. That is, the generated states' properties should be consistent with the ground truth dataset, both in the guided and unguided settings.

We design set of tests to assess the coherence to the ground truth dataset of the output trajectories across following dimensions:

- temporally (transitions across the same variable should be smooth)
- spatially (region of interest)
- vertically (pressure levels)
- across variables (patterns should be coherent to the learned weather transition function)

We will benchmark these properties against the ground-truth data (anchor) and measure the difference between unguided, guided-unguided, and guided states. For the first three concerns we compute difference maps in the axis of interest and measure min, mean, max, and std across maps. For the last property we measure all the cross-variable distributions over the weather steps of the ground truth data, and then report the deviation from the average of the unguided, guided-unguided, and guided states.

For all four tests we average over different experiments and report the avg and std over all runs. This will hopefully give us an indication of whether the generated extremes are faithful to the data distribution.

(Remember, in order for this to work the base-forecast have to have missed potential heat spots –> the extreme is not only about the average, wich the forecast will most probably cover faithfully, but about the extreme spots (max). Both of to be tested anyway.)

Here a visualization summarizing these tests: — commented out for speed of compilation

In the next chapter we will expose the settings for the final experiments, and present the relevant results, and then proceed to discuss them in the following chapter.

Chapter 6

Experiments

6.1 Flow time evaluation

6.2 Weather time evaluation

6.3 Use cases

For final use cases we then fix most of the aformentioned parameters and increase N and M, producing the experiments we had in mind when setting the goals of the project.

6.3.1 Europa 2020-06-01

6.3.2 Australia 2020-01-01

6.3.3 North America 2020-07-01

Chapter 7

Discussion

The algorithm doesn't produce diverse results, because the gradients are really stable. This has to do with the fact that we have learned one and only one flow model (transition function). If we had multiple residual diffusion model, we would expect the possible effects to be more diverse. However, the second interpretation might be that the gradient provides with the information about the most likely direction of change (which is true under the learned data distribution).

This happens because the guidance vector is a scaled version of the flow model's gradient wrt to the noisy state. The gradient is stable across the full trajectory, because the only variable inputs are the noise itself and the timestep of the flow. The stochasticity of the noise is not enough to produce diverse results. The direction of the gradient is always the same. If we say "let the masked average go up by phi percent", the only thing that changes is the magnitude of the gradient, but not its direction, which is specified by the Jacobian.

This is in contrast to guidance settings.

Shift between atmosphere and stratosphere.

Put here all observations

Chapter 8

Conclusion

8.1 Summary

8.2 Future Directions

8.3 Conclusive Remarks

Appendices

Appendix A

Code-base structure

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Eidesstattliche Erklärung

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